$\int_0^{\frac{\pi}{2}} \frac{\sin x \cos x}{\sqrt{1+\sin^2 x}} dx$

$u := 1 + \sin^2 x \quad du = 2 \sin x \cos x$

$\frac{1}{2} \int_1^2 \frac{1}{\sqrt{u}} du$

$\sqrt{u} \ |_1^2$

$\sqrt{2} - 1$